USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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Country: Zip/Postal Code:

Phone: FAX:

Name: Email:

Company: Mailstop:

Address:

City: State/Province:

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: tSrcSettle_deprecation
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
Modification to align to ECRtSrcTransition: https://groups.usb.org/wg/powerdelivery/document/18788

Benefits as a result of the proposed changes:
Alignment to main specification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None if DUT properly implemented

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.PS.SRC.1.Multiple Request Load Test

i. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* sent in responds to the *Accept* Message. The Tester performs the following checks:

1. The Tester checks that V_{BUS} remains within *vSrcNew, vAvsNew* or *vPpsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SRC.1#3]
2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. (See **Error! Reference source not found.**) [TEST.PD.PS.SRC.1#4]
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcSettle* max. [TEST.PD.PS.SRC.1#5]
4. The Tester checks that the V_{BUS} voltage remains within *vSrcNew, vAvsNew* or *vPpsNew* limits between *tSrcSettle* max and 100ms after *tSrcSettle* max. [TEST.PD.PS.SRC.1#6]

TO:

TEST.PD.PS.SRC.1.Multiple Request Load Test

i. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* sent in responds to the *Accept* Message. The Tester performs the following checks:

5. The Tester checks that V_{BUS} remains within *vSrcNew, vAvsNew* or *vPpsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SRC.1#3]
6. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. (See **Error! Reference source not found.**) [TEST.PD.PS.SRC.1#4]
7. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SRC.1#5]
8. The Tester checks that the V_{BUS} voltage remains within *vSrcNew, vAvsNew* or *vPpsNew* limits between ~~*tSrcSettle*~~ tSrcReady and 100ms after ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SRC.1#6]

USB PD CTS ENGINEERING CHANGE NOTICE

FROM:

TEST.PD.PS.SRC.2: PDO Transition

d. The Tester performs the following checks:
   i. For transitions to a new PDO: Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:
      1. The Tester checks the V_{BUS} is within *vSrcNew* or *vPpsNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SRC.2#3]
      2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.SRC.2#4] (see **Error! Reference source not found.**)
      3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcSettle* max. [TEST.PD.PS.SRC.2#5]
      4. The Tester checks that the V_{BUS} voltage is within *vSrcNew*, *vPpsNew* or *vAvsNew* limits between *tSrcSettle* max and 100ms after *tSrcSettle* max. [TEST.PD.PS.SRC.2#6]
      5. The Tester checks that the UUT does not send *PS_RDY* before the V_{BUS} is within *vSrcNew*, *vPpsNew* or *vAvsNew*. [TEST.PD.PS.SRC.2#7]

TO:

TEST.PD.PS.SRC.2: PDO Transition

g. The Tester performs the following checks:
   d. For transitions to a new PDO: Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:
      d. The Tester checks the V_{BUS} is within *vSrcNew* or *vPpsNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SRC.2#3]
      e. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.SRC.2#4] (see **Error! Reference source not found.**)
      f. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SRC.2#5]
      g. The Tester checks that the V_{BUS} voltage is within *vSrcNew*, *vPpsNew* or *vAvsNew* limits between ~~*tSrcSettle*~~ tSrcReady max and 100ms after ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SRC.2#6]
      h. The Tester checks that the UUT does not send *PS_RDY* before the V_{BUS} is within *vSrcNew*, *vPpsNew* or *vAvsNew*. [TEST.PD.PS.SRC.2#7]

FROM:

TEST.PD.PS.SNK.2: Initial Sink PDO Transitions Post PR Swap

d. The Tester performs the following checks:
Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:
   1. The Tester checks the V_{BUS} is within *vSrcNew* or *vPpsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SNK.2#6]

2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.SNK.2#7] (see **Error! Reference source not found.**)
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits upon crossing into the *vSrcValid* limits until *tSrcSettle* max. [TEST.PD.PS.SNK.2#8]
4. The Tester checks that the V_{BUS} voltage is within *vSrcNew* or *vPpsNew* limits between *tSrcSettle* max and 100ms after *tSrcSettle* max. [TEST.PD.PS.SNK.2#9]
5. The Tester checks that the UUT does not send *PS_RDY* before the V_{BUS} is within *vSrcNew* or *vPpsNew*. [TEST.PD.PS.SNK.2#10]

TO:

TEST.PD.PS.SNK.2: Initial Sink PDO Transitions Post PR Swap

d. The Tester performs the following checks:

Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:

1. The Tester checks the V_{BUS} is within *vSrcNew* or *vPpsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SNK.2#6]
2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.SNK.2#7] (see **Error! Reference source not found.**)
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits upon crossing into the *vSrcValid* limits until ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SNK.2#8]
4. The Tester checks that the V_{BUS} voltage is within *vSrcNew* or *vPpsNew* limits between ~~*tSrcSettle*~~ tSrcReady max and 100ms after ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SNK.2#9]
5. The Tester checks that the UUT does not send *PS_RDY* before the V_{BUS} is within *vSrcNew* or *vPpsNew*. [TEST.PD.PS.SNK.2#10]

FROM:

TEST.PD.PS.SNK.3 Multiple Request Load Test Post PR Swap

iii. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* is sent in response to the *Accept* Message. The Tester performs the following checks:

1. The Tester checks that V_{BUS} remains within *vSrcNew, vAvsNew* or *vPpsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SNK.3#6]
2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.SNK.3#7] (see Error! Reference source not found.)
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcSettle* max. [TEST.PD.PS.SNK.3#8]
4. The Tester checks that the V_{BUS} voltage remains within *vSrcNew, vAvsNew* or *vPpsNew* limits between *tSrcSettle* max and 100ms after *tSrcSettle* max. [TEST.PD.PS.SNK.3#9]

To:

TEST.PD.PS.SNK.3 Multiple Request Load Test Post PR Swap

iv. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* is sent in response to the *Accept* Message. The Tester performs the following checks:

5. The Tester checks that V_{BUS} remains within *vSrcNew, vAvsNew* or *vPpsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.SNK.3#6]
6. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.SNK.3#7] (see Error! Reference source not found.)
7. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SNK.3#8]
8. The Tester checks that the V_{BUS} voltage remains within *vSrcNew, vAvsNew* or *vPpsNew* limits between ~~*tSrcSettle*~~ tSrcReady max and 100ms after ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.SNK.3#9]

FROM:

TEST.PD.PS.EPR.SRC3.1: Multiple EPR Request Load Test

e. The Tester performs the following checks:
   i. If the transition involves a current decrease (but no PDO change), the Tester decreases current to the new value within *tSnkNewPower* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#3]
   ii. If the transition involves a current increase (but no PDO change), the Tester increases current to the new value after the last bit of the *GoodCRC* sent in response to the *PS_RDY* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#4]
   iii. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:
      1. The Tester checks that V_{BUS} remains within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.1#5]
      2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.1#6]. (See Error! Reference source not found.)
      3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcSettle* max. [TEST.PD.PS.EPR.SRC3.1#7]
      4. The Tester checks that the V_{BUS} voltage remains within *vSrcNew* or *vAvsNew* limits between *tSrcSettle* max and 100ms after *tSrcSettle* max. [TEST.PD.PS.EPR.SRC3.1#8]

To:

TEST.PD.PS.EPR.SRC3.1: Multiple EPR Request Load Test

e. The Tester performs the following checks:
   i. If the transition involves a current decrease (but no PDO change), the Tester decreases current to the new value within *tSnkNewPower* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#3]
   ii. If the transition involves a current increase (but no PDO change), the Tester increases current to the new value after the last bit of the *GoodCRC* sent in response to the *PS_RDY* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#4]

iii. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:

1. The Tester checks that V_{BUS} remains within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.1#5]
2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.1#6]. (See Error! Reference source not found.)
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.EPR.SRC3.1#7]
4. The Tester checks that the V_{BUS} voltage remains within *vSrcNew* or *vAvsNew* limits between ~~*tSrcSettle*~~ tSrcReady max and 100ms after ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.EPR.SRC3.1#8]

FROM:

TEST.PD.PS.EPR.SRC3.2: PDO Transitions in EPR Mode

e. The Tester performs the following checks:
Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:

1. The Tester checks the V_{BUS} is within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.2#5]
2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.2#6] (See Error! Reference source not found.)
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcSettle* max. [TEST.PD.PS.EPR.SRC3.2#7]
4. The Tester checks that the V_{BUS} voltage is within *vSrcNew* or *vAvsNew* limits between *tSrcSettle* max and 100ms after *tSrcSettle* max. [TEST.PD.PS.EPR.SRC3.2#8]

TO:

TEST.PD.PS.EPR.SRC3.2: PDO Transitions in EPR Mode

f. The Tester performs the following checks:
Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:

5. The Tester checks the V_{BUS} is within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.2#5]
6. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.2#6] (See Error! Reference source not found.)

7. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.EPR.SRC3.2#7]
8. The Tester checks that the V_{BUS} voltage is within *vSrcNew* or *vAvsNew* limits between ~~*tSrcSettle*~~ tSrcReady max and 100ms after ~~*tSrcSettle*~~ tSrcReady max. [TEST.PD.PS.EPR.SRC3.2#8]